

MATH120-24S2 Discrete Mathematics

Tutorial 10

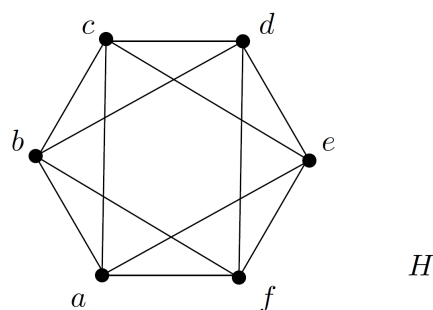
References

- Topics 17 – 18: Eulerian and Hamiltonian Graphs; Counting

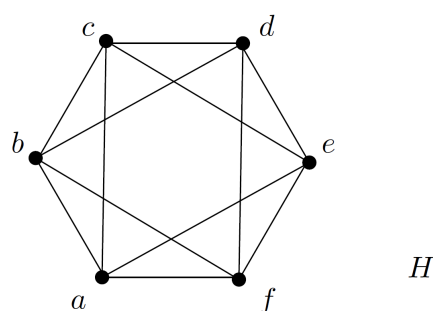
Work on the following problems during the tutorial in week 11.

- ⇒ 1. [B] Recall Theorem 2.1 in the lecture notes which says that a connected graph G is Eulerian if and only if all vertices of G are of even degree. The inductive proof of Theorem 2.1 provides the following algorithm for constructing an Eulerian circuit of an Eulerian graph. (You should check why this is so.)
- Algorithm:** EULERCIRCUIT(G, v)
- Input:** An Eulerian graph $G = (V, E)$ and a vertex v of G .
- Output:** An Eulerian circuit of G that begins and ends at v .
1. Starting at v , construct a closed walk W of G in which no edge is included more than once.
 2. Let $G' = (V, E - E(W))$, where $E(W)$ denotes the set of edges of W .
 3. Let $C_1, C_2, \dots, C_k$ denote the components of G' . For all $i \in \{1, 2, \dots, k\}$, choose a vertex $v_i \in V(C_i) \cap V(W)$, where $V(C_i)$ and $V(W)$ denote the set of vertices of C_i and W , respectively.
 4. For each $i \in \{1, 2, \dots, k\}$, call EULERCIRCUIT(C_i, v_i). Let $W_1, \dots, W_k$ be the resulting Eulerian circuits.
 5. Output the closed walk that is obtained from W by replacing, for all i , v_i with W_i .

Using EULERCIRCUIT, find an Eulerian circuit of the following graph H . Include all intermediate steps.



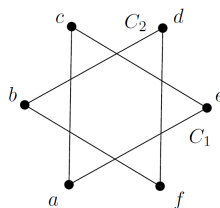
Solution:



By Theorem 2.1, H is Eulerian. Using `EULERCIRCUIT`, we can construct an Eulerian circuit of H starting at a as follows.

Step 1 Construct a closed walk W starting at a so that no edge is included twice, $W = abcdefa$ say.

Step 2 Removing the edges in W from H , we obtain the following graph that consists of two components C_1 and C_2 .



H with the edges in W removed.

Choose the vertex a in $V(C_1) \cap W$ and the vertex d in $V(C_2) \cap W$.

Step 3 Calling `EULERCIRCUIT` twice, we construct an Eulerian circuit for C_1 starting at a , $W_1 = acea$ say, and an Eulerian circuit for C_2 starting at d , $W_2 = dfbd$ say.

Step 4 Replacing a with W_1 and b with W_2 in W , we output $aceabdfbcdefa$ (an Eulerian circuit of H starting at a).

2. [M] Let G be a connected graph that is not Eulerian. Show that it is always possible to add to G a single vertex together with some edges each of which joins this new vertex to a vertex of G so that the resulting graph is Eulerian.

Solution:

Let G' be the graph obtained from G by adding a new vertex v and, for each vertex u of G of odd degree, adding an edge joining v and u . Observe that every vertex of $V(G') - \{v\}$ has even degree. Furthermore, by Corollary 1.2, G has an even number of vertices of odd degree, and so v also has even degree. It now follows by Theorem 2.1 that G' is Eulerian.

3. [B] A Graph G has 7 vertices of degrees 4, 4, 5, 5, 5, 6 and 7. Determine whether G is Hamiltonian, not Hamiltonian, or if you don't have enough information.

Solution:

G is Hamiltonian because it satisfies the conditions of Dirac's theorem. Each vertex has degree at least $7/2$.

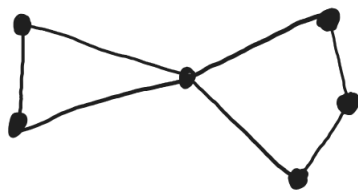
$\Rightarrow$

4. [M]
- Give an example of a graph on 6 vertices that is Hamiltonian but not Eulerian. Justify why it is not Eulerian.
 - Give an example of a graph on 6 vertices that is Eulerian but not Hamiltonian. Justify why it is not Hamiltonian.

Solution:

- One example is a cycle on 6 vertices with one extra edge added between any two distinct vertices. We started with a cycle so we know it is Hamiltonian. Then adding one extra edge means two vertices now have odd degree so it is not Eulerian.

(b) One example is the following graph.



Every vertex has even degree so it is Eulerian.

It is not Hamiltonian. Recall that a Hamilton cycle in a graph uses exactly two edges incident with each vertex. In particular, any potential Hamiltonian cycle would use two edges incident with v . Suppose it uses uv and vw , then it must also use uw and hence is not a Hamilton cycle. Next suppose it uses uv and vx , then it must also use xy . However it must use another edge incident with y but cannot use yv , a contradiction.

⇒ 5. [M] Find conditions for n and m so that $K_{n,m}$ is Hamiltonian.

Solution:

If $n = m \geq 2$ then Dirac's Theorem is satisfied because the degree of each vertex is n and there are $2n$ vertices in the graph, so $K_{n,n}$ is Hamiltonian when $n \geq 2$.

Otherwise, $K_{n,m}$ is not Hamiltonian. First note that $K_{1,1}$ is a single edge so is not Hamiltonian. Then suppose $n < m$, then any cycle in $K_{n,m}$ has at most $2n$ vertices and therefore is not a Hamiltonian cycle (the total number of vertices is $n + m > 2n$).

⇒ 6. A team has 13 members.

(a) [B] How many ways can a group of seven be chosen to work on a project?

(b) [M] Suppose seven team members are women and six are men.

- i. How many groups of seven can be chosen that contain four women and three men?
- ii. How many groups of seven can be chosen that contain at least one man?

Solution:

(a) $\binom{13}{7}$

(b) i. $\binom{7}{4} \times \binom{6}{3}$

ii. $\binom{13}{7} - 1$. Note that there is only one way to choose a group of 7 with no men so all remaining groups must contain at least one man.

$\Rightarrow$ 7. [B] There are 20 different types of inputs to a computer programme. In how many ways can 8 of them be selected if

(a) order does not matter, and

(b) order matters.

Solution:

(a) $\binom{20}{8} = \frac{20!}{8!(20-8)!} = \frac{20!}{8! \cdot 12!}$

(b) $20 \cdot 19 \cdot 18 \cdot 17 \cdot 16 \cdot 15 \cdot 14 \cdot 13$

8. [M] Let n be a non-negative integer. Show that

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n-1} + \binom{n}{n} = 2^n.$$

Hint: There is a quick way to do it!

Solution:

For each $r \in \{0, 1, 2, \dots, n\}$, the binomial coefficient $\binom{n}{r}$ is the number of r -element subsets of a set of size n . Call the latter set S . Therefore, the left-hand-side is the number of 0-element subsets of S plus the number of 1-element subsets S plus the number of 2-element subsets of S and so on. In particular, the left-hand-side is the total number of subsets of a S . This is precisely the power set of the set S . The power set of S has exactly $2^{|S|}$ elements as stated in the lecture notes.

Why is that true? Let $A = \{a_1, a_2, \dots, a_n\}$ be an n -element set. What is the total number of subsets of A ? For each subset, a_1 is in it or it is not, a_2 is in it or it is not, and so on. Therefore, by the multiplication rule, the total number of number of subsets of A is

$$2 \times 2 \times \cdots \times 2 = 2^n.$$

Hence the total number of subsets of a set of size n is 2^n .

Alternatively, you might have encountered the binomial theorem before:

$$(a + b)^n = \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i.$$

Plugging in $a = b = 1$ yields the expression we needed to prove.

9. [M]

- (a) How many four-digit alpha-numeric PINs are there? I.e. a sequence of any four symbols chosen from 26 letters of the alphabet or ten digits.
- (b) how many PINs are there if we exclude PINs that repeat one symbol four times? E.g. AAAA or 3333
- (c) What if we also exclude four consecutive digits, e.g. 0123 or BCDE? First discuss exactly which combinations you will exclude.

Solution:

- (a) There are 36 symbols, so 36^4 different PINs.
- (b) There are 36 PINs consisting of four repeated symbols and we want to exclude these. So there are $36^4 - 36$ such PINs.
- (c) First we count how many PINS we want to *exclude*. We exclude 0123, 1234,..., 6789 (there are 7 PINs in this list) and we also exclude ABCD, ..., WXYZ (there are 23 PINs in this list). So, also excluding the PINs from (b) we have $36^4 - 36 - 7 - 23 = 36^4 - 66$.